

Pearl Rana

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Expected Graduation: 2028  [LinkedIn](#)  [Portfolio](#)

EDUCATION

•Thapar Institute of Engineering and Technology, Patiala

Bachelor of Engineering (B.E.) in Computer Science and Business Systems

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PROJECTS

•Pigment Demand Studio

 [GitHub](#)

React.js, FastAPI, SQLite, ARIMA

- Built a full-stack demand forecasting platform with a React.js front end integrated with FastAPI backend services for real-time data exchange.
- Engineered RESTful API endpoints to connect ARIMA-based forecasting models with the application, enabling automated demand and inventory predictions.
- Designed interactive dashboards to visualize inventory trends and forecasting insights, improving data accessibility and usability.

•ServeStats – Restaurant Data Analytics System

 [GitHub](#)

SQL, DBMS, React.js

- Wrote optimized SQL queries using JOIN, GROUP BY, HAVING, and aggregate functions to analyze restaurant sales and customer data.
- Applied database normalization principles to improve data integrity, reduce redundancy, and streamline query performance.
- Built interactive React.js dashboards to visualize revenue trends, customer behavior, and product performance for data-driven decisions.

•Personal Portfolio Website

Live

HTML, CSS, GitHub Pages

- Designed and customized a modern personal portfolio website, integrating multiple sections such as projects, skills, and contact details into a cohesive user experience.
- Ensured responsiveness across devices and optimized navigation for smooth, intuitive browsing on both desktop and mobile.
- Maintained a clean, organized project structure using Git and GitHub for version control, and successfully deployed the site using Vercel.

TECHNICAL SKILLS

Languages: Python, C++, C, JavaScript, SQL

Frameworks: React.js, FastAPI

Web: HTML5, CSS3

Databases: SQLite, MySQL

Core CS: Data Structures and Algorithms, Object Oriented Programming, Operating Systems, DBMS

Tools: Git, GitHub, VS Code

Relevant Coursework: Software Engineering, Design Thinking, Linear Algebra, Probability and Statistics, Data Analytics

LEADERSHIP & ACTIVITIES

•R&D and Tech Team Member – Echoes

- Collaborated with cross-functional teams to plan and execute technical events and support event operations.
- Coordinated logistics and technical resources for workshops and competitions, strengthening organizational and teamwork skills.

•Member – ACM Student Chapter

- Participated in coding workshops and peer-learning sessions to strengthen programming and problem-solving skills.
- Engaged with technical talks and collaborative coding sessions to stay current with industry practices and tools.

•Member – Mudra Cultural Society

- Assisted in organizing cultural events and collaborated with teams to ensure smooth execution of activities.
- Coordinated with peers under tight timelines to deliver well-planned events, building communication and leadership skills.